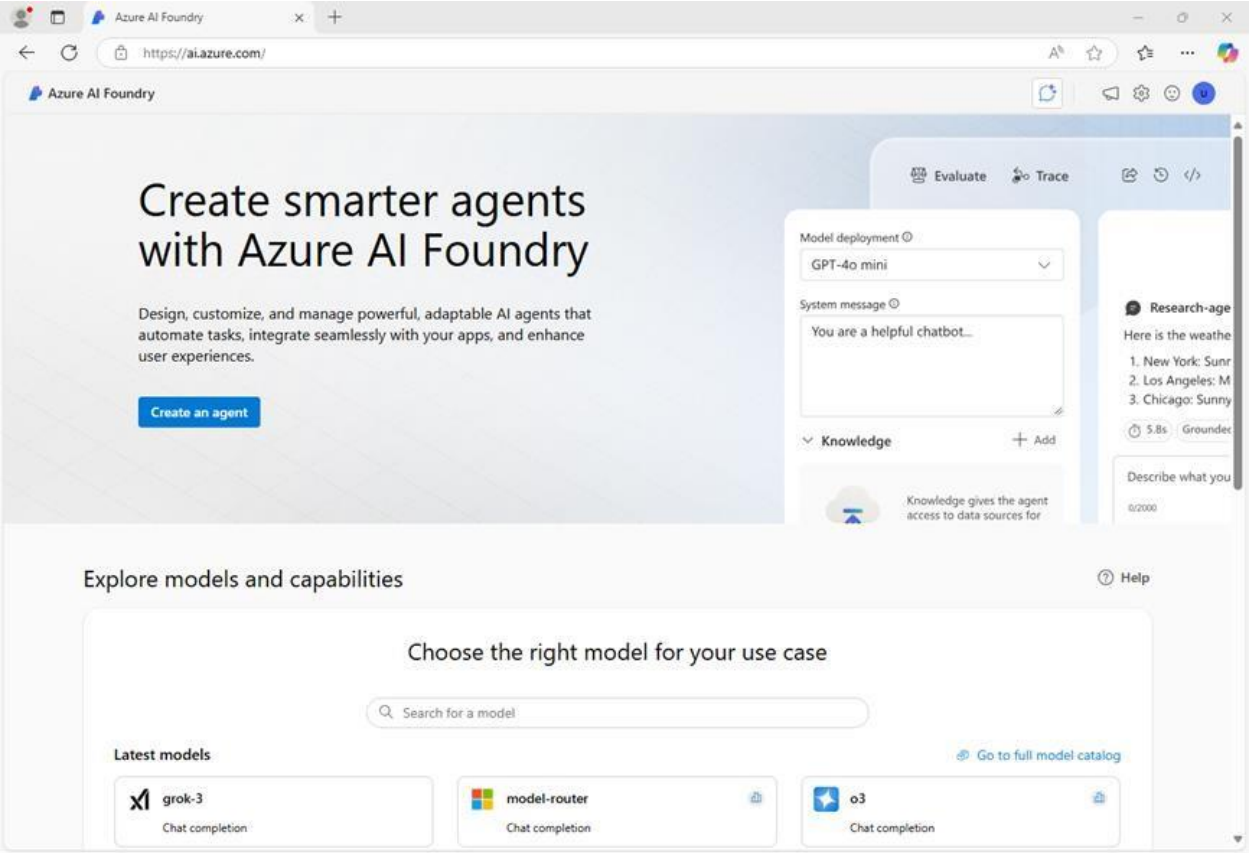
Use a prompt flow to manage conversation in a chat app

In this exercise, you'll use Microsoft Foundry portal's prompt flow to create a custom chat app that uses a user prompt and chat history as inputs, and uses a GPT model from Azure OpenAI to generate an output.

Create a Foundry hub and project (Please check with your instructor if Foundry hub, project and model are deployed, if yes, feel free to jump to the next section in the same doc that is - 'Create a prompt flow')

The features of Foundry we're going to use in this exercise require a project that is based on a Foundry *hub* resource.

1. In a web browser, open the Foundry portal at `https://ai.azure.com` and sign in using your Azure credentials. Close any tips or quick start panes that are opened the first time you sign in, and if necessary use the **Foundry** logo at the top left to navigate to the home page, which looks similar to the following image (close the **Help** pane if it's open):



2. In the browser, navigate to `https://ai.azure.com/managementCenter/allResources` and select **Create new**. Then choose the option to create a new **AI hub resource**.
3. In the **Create a project** wizard, enter a valid name for your project, and select the option to create a new hub. Then use the **Rename hub** link to specify a valid name for your new hub, expand **Advanced options**, and specify the following settings for your project:
    a. **Subscription**: *Your Azure subscription*
    b. **Resource group**: *Create or select a resource group*
    c. **Region**: East US 2 or Sweden Central (*In the event of a quota limit being exceeded later in the exercise, you may need to create another resource in a different region.*)

Note: If you're working in an Azure subscription in which policies are used to restrict allowable resource names, you may need to use the link at the bottom of the **Create a new project** dialog box to create the hub using the Azure portal.

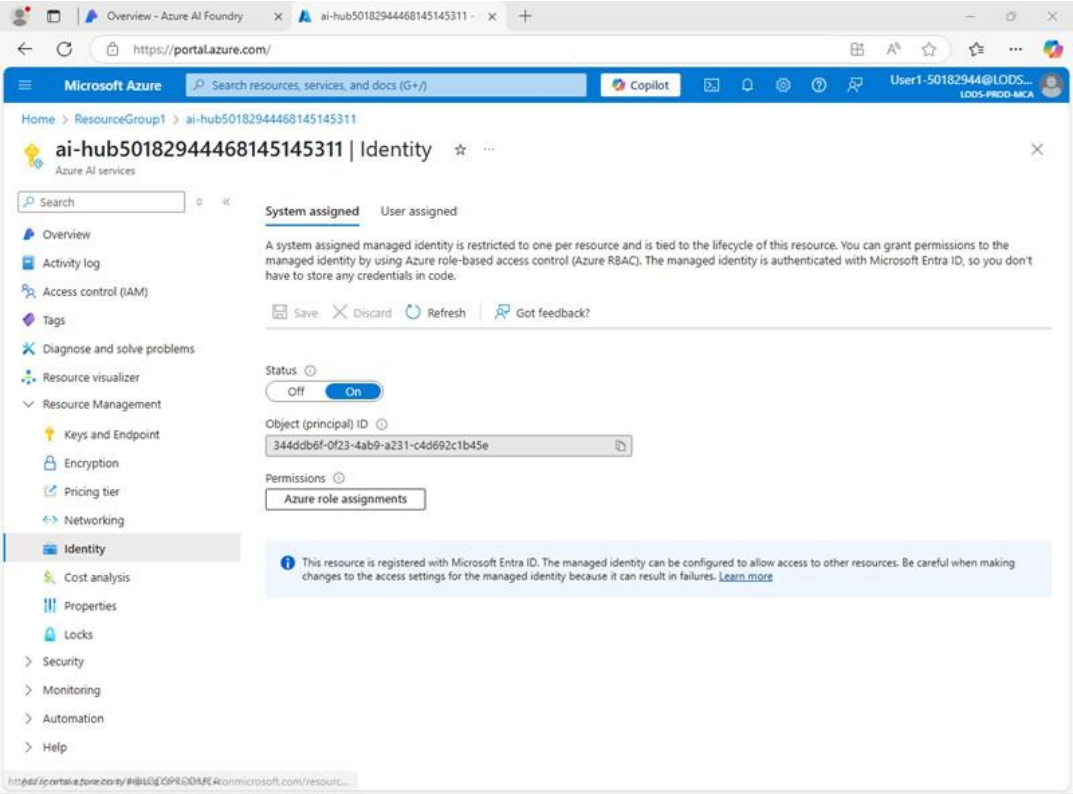
Tip: If the **Create** button is still disabled, be sure to rename your hub to a unique alphanumeric value.

4. Wait for your project to be created.

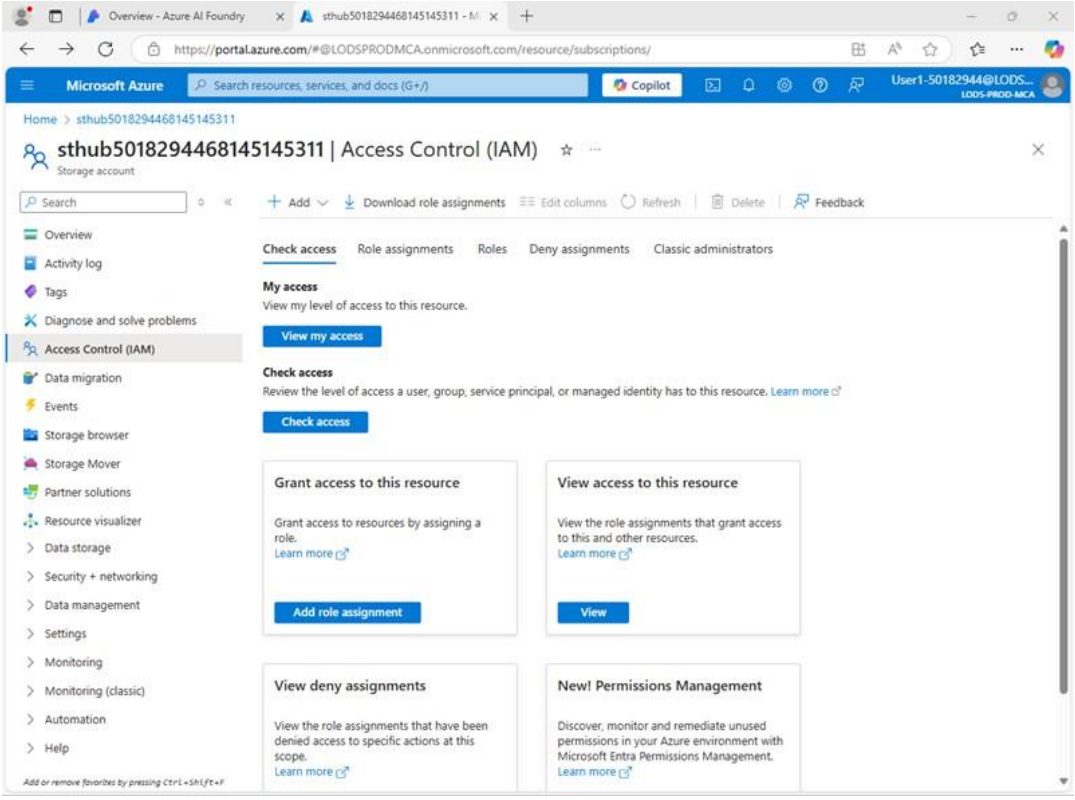
Configure resource authorization

The prompt flow tools in Foundry create file-based assets that define the prompt flow in a folder in blob storage. Before exploring prompt flow, let's ensure that your Foundry resource has the required access to the blob store so it can read them.

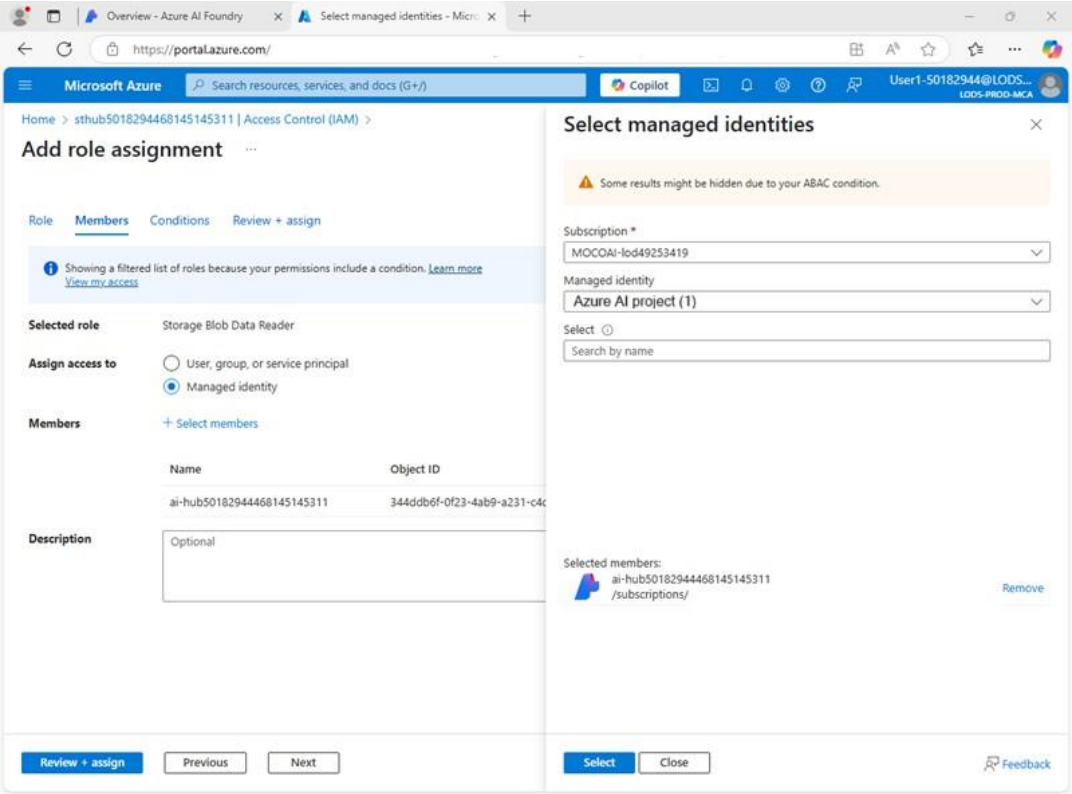
1. In a new browser tab, open the Azure portal at `https://portal.azure.com`, signing in with your Azure credentials if prompted; and view the resource group containing your Azure AI hub resources.
2. Select the **Foundry** resource for your hub to open it. Then expand its **Resource Management** section and select the **Identity** page:



3. If the status of the system assigned identity is **Off**, switch it **On** and save your changes. Then wait for the change of status to be confirmed.
4. Return to the resource group page, and then select the **Storage account** resource for your hub and view its **Access Control (IAM)** page:



5. Add a role assignment to the `Storage blob data reader` role for the managed identity used by your Foundry project resource:



6. When you've reviewed and assigned the role access to allow the Foundry managed identity to read blobs in the storage account, close the Azure portal tab and return to the Foundry portal.

Deploy a generative AI model

Now you're ready to deploy a generative AI language model to support your prompt flow application.

1. In the pane on the left for your project, in the **My assets** section, select the **Models + endpoints** page.
2. In the **Models + endpoints** page, in the **Model deployments** tab, in the **+ Deploy model** menu, select **Deploy base model**.
3. Search for the **gpt-4o** model in the list, and then select and confirm it.
4. Deploy the model with the following settings by selecting **Customize** in the deployment details:
    a. **Deployment name**: *A valid name for your model deployment*
    b. **Deployment type**: Global Standard
    c. **Automatic version update**: Enabled

   d. **Model version**: *Select the most recent available version*
   e. **Connected AI resource**: *Select your Azure OpenAI resource connection*
   f. **Tokens per Minute Rate Limit (thousands)**: 50K *(or the maximum available in your subscription if less than 50K)*
   g. **Content filter**: DefaultV2

Note: Reducing the TPM helps avoid over-using the quota available in the subscription you are using. 50,000 TPM should be sufficient for the data used in this exercise. If your available quota is lower than this, you will be able to complete the exercise but you may experience errors if the rate limit is exceeded.

5. Wait for the deployment to complete.

Create a prompt flow

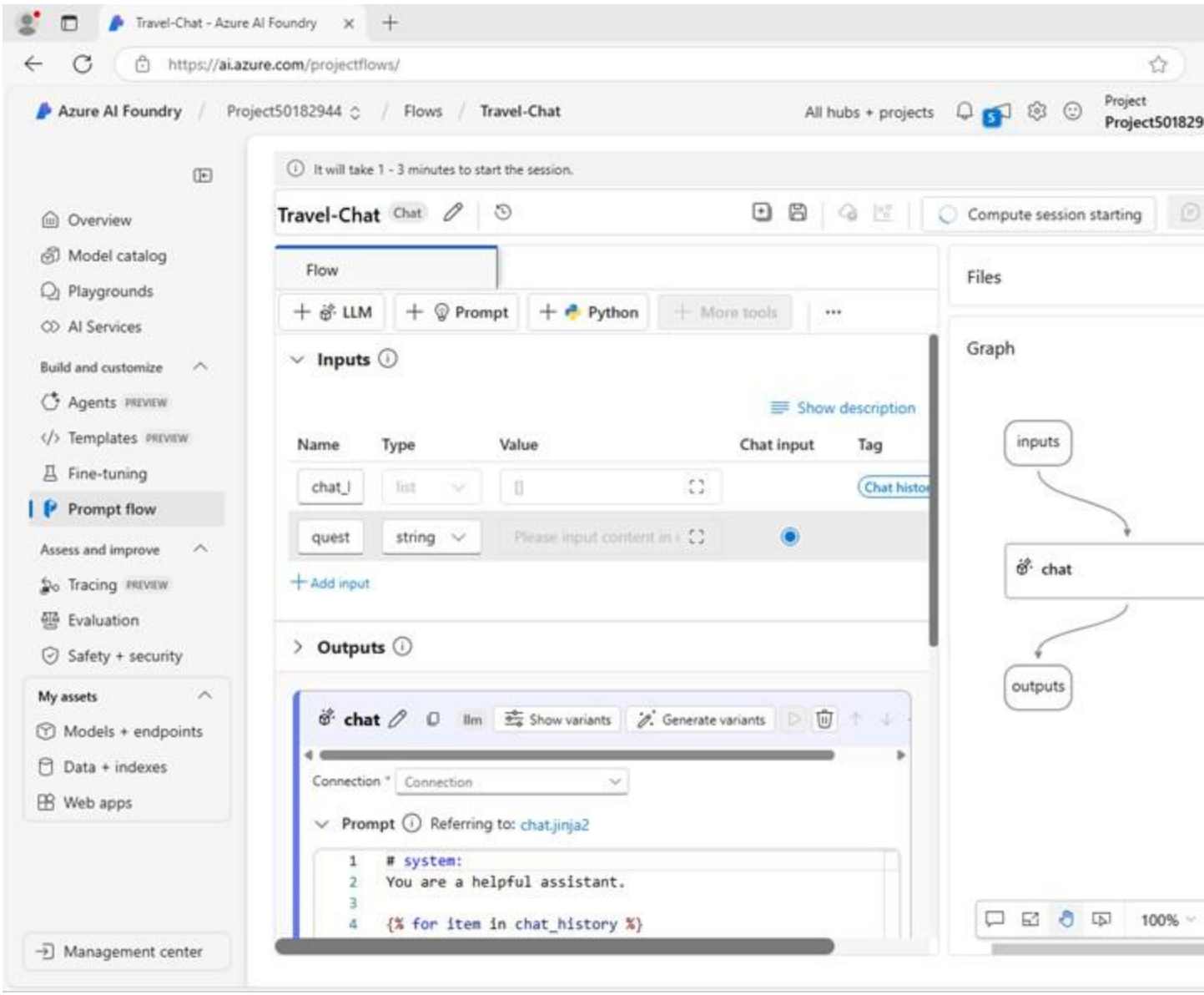
A prompt flow provides a way to orchestrate prompts and other activities to define an interaction with a generative AI model. In this exercise, you'll use a template to create a basic chat flow for an AI assistant in a travel agency.

1. In the Foundry portal navigation bar, in the **Build and customize** section, select **Prompt flow**.
2. Create a new flow based on the **Chat flow** template, specifying `Travel-Chat` as the folder name.

A simple chat flow is created for you.

Tip: If a permissions error occurs. Wait a few minutes and try again, specifying a different flow name if necessary.

3. To be able to test your flow, you need compute, and it can take a while to start; so select **Start compute session** to get it started while you explore and modify the default flow.
4. View the prompt flow, which consists of a series of *inputs*, *outputs*, and *tools*. You can expand and edit the properties of these objects in the editing panes on the left, and view the overall flow as a graph on the right:



5. View the **Inputs** pane, and note that there are two inputs (chat history and the user's question)
6. View the **Outputs** pane and note that there's an output to reflect the model's answer.
7. View the **Chat** LLM tool pane, which contains the information needed to submit a prompt to the model.
8. In the **Chat** LLM tool pane, for **Connection**, select the connection for the Azure OpenAI service resource in your AI hub. Then configure the following connection properties:
    a. **Api**: chat

   b. **deployment_name**: *The gpt-4o model you deployed*
   c. **response_format**: {"type":"text"}

9. Modify the **Prompt** field as follows, removing the \ escape in the YAML loop:

```
# system:
**Objective**: Assist users with travel-related inquiries, offering tips, advice, and
recommendations as a knowledgeable travel agent.

**Capabilities**:
- Provide up-to-date travel information, including destinations, accommodations,
transportation, and local attractions.
- Offer personalized travel suggestions based on user preferences, budget, and travel
dates.
- Share tips on packing, safety, and navigating travel disruptions.
- Help with itinerary planning, including optimal routes and must-see landmarks.
- Answer common travel questions and provide solutions to potential travel issues.

**Instructions**:
1. Engage with the user in a friendly and professional manner, as a travel agent
would.
2. Use available resources to provide accurate and relevant travel information.
3. Tailor responses to the user's specific travel needs and interests.
4. Ensure recommendations are practical and consider the user's safety and comfort.
5. Encourage the user to ask follow-up questions for further assistance.

\{\% for item in chat_history \%\}
# user:
\{\{item.inputs.question\}\}
# assistant:
\{\{item.outputs.answer\}\}
\{\% endfor \%\}

# user:
\{\{question\}\}
```

Read the prompt you added so you are familiar with it. It consists of a system message (which includes an objective, a definition of its capabilities, and some instructions), and the chat history (ordered to show each user question input and each previous assistant answer output)

10. In the **Inputs** section for the **Chat** LLM tool (under the prompt), ensure the following variables are set:
    a. **question** (string): ${inputs.question}

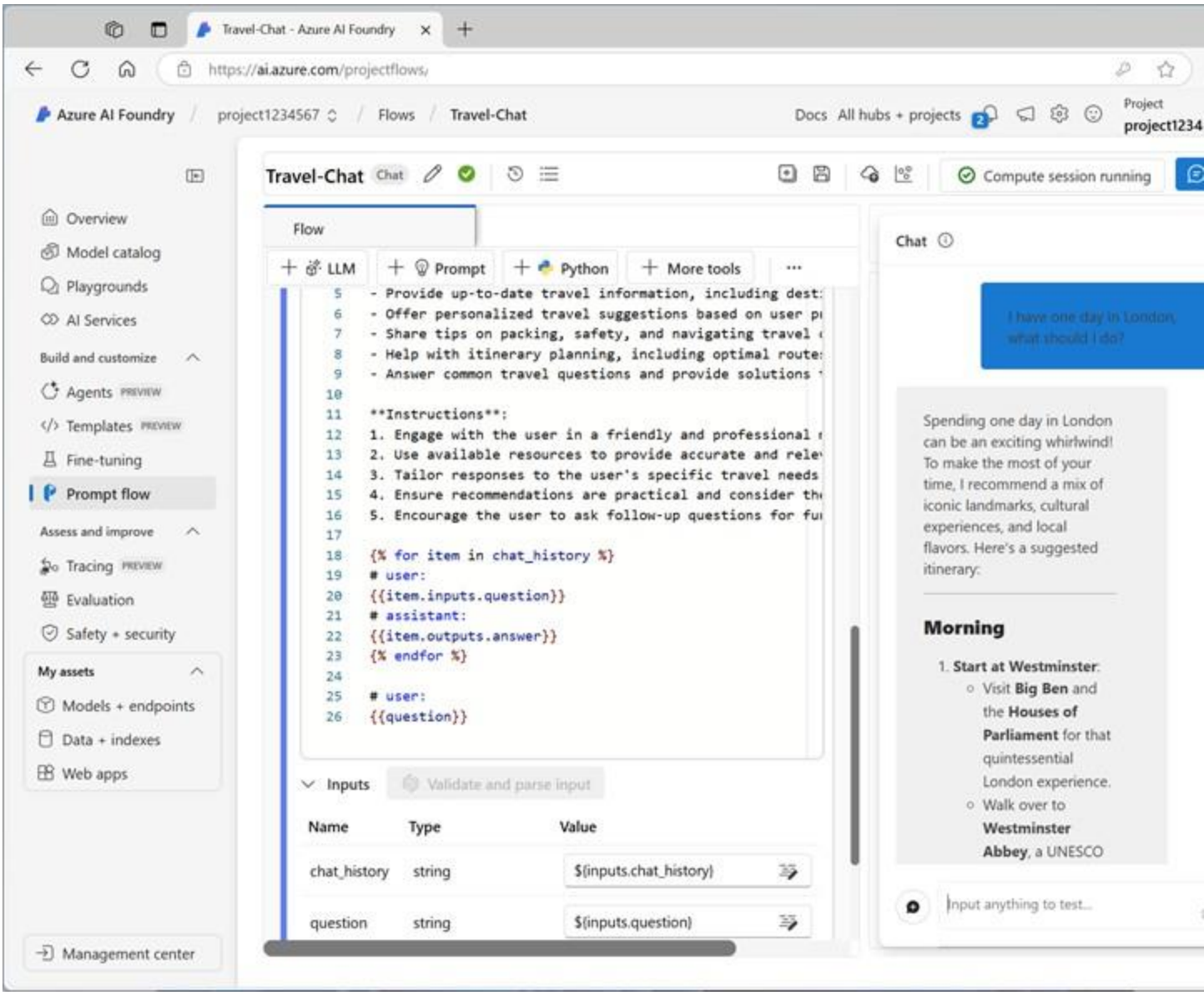
    b. **chat_history** (string): ${inputs.chat_history}
11. Save the changes to the flow.

Note: In this exercise, we'll stick to a simple chat flow, but note that the prompt flow editor includes many other tools that you could add to the flow, enabling you to create complex logic to orchstrate conversations.

Test the flow

Now that you've developed the flow, you can use the chat window to test it.

1. Ensure the compute session is running. If not, wait for it to start.
2. On the toolbar, select **Chat** to open the **Chat** pane, and wait for the chat to initialize.
3. Enter the query: `I have one day in London, what should I do?` and review the output. The Chat pane should look similar to this:


Travel-Chat - Azure AI Foundry
https://ai.azure.com/projectflows/
Azure AI Foundry / project1234567 / Flows / Travel-Chat
Docs All hubs + projects
Project
project1234
Travel-Chat Chat
Compute session running
Overview
Model catalog
Playgrounds
AI Services
Build and customize
Agents PREVIEW
Templates PREVIEW
Fine-tuning
Prompt flow
Assess and improve
Tracing PREVIEW
Evaluation
Safety + security
My assets
Models + endpoints
Data + indexes
Web apps
Management center
Flow
LLM
Prompt
Python
More tools
5 - Provide up-to-date travel information, including dest
6 - Offer personalized travel suggestions based on user p
7 - Share tips on packing, safety, and navigating travel
8 - Help with itinerary planning, including optimal route
9 - Answer common travel questions and provide solutions
10
11 **Instructions**:
12 1. Engage with the user in a friendly and professional
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15 4. Ensure recommendations are practical and consider th
16 5. Encourage the user to ask follow-up questions for fu
17
18 {% for item in chat_history %}
19 # user:
20 {{item.inputs.question}}
21 # assistant:
22 {{item.outputs.answer}}
23 {% endfor %}
24
25 # user:
26 {{question}}
Inputs
Validate and parse input
Name
Type
Value
chat_history
string
${inputs.chat_history}
question
string
${inputs.question}
Chat
I have one day in London, what should I do?
Spending one day in London can be an exciting whirlwind! To make the most of your time, I recommend a mix of iconic landmarks, cultural experiences, and local flavors. Here's a suggested itinerary:
Morning
1. Start at Westminster:
Visit Big Ben and the Houses of Parliament for that quintessential London experience.
Walk over to Westminster Abbey, a UNESCO
Input anything to test...